Week 5 question sheet

Problem sheet

Instructions

Attempt all ten questions and all 17 scorable parts. The target time is 240 minutes and the total is 100 marks. No calculator or software is needed. Show every regularity, nonsingularity, domain or support condition needed for your method. Gradients and vectors are columns. Give exact values.

Use hints.md only after a serious attempt. Do not consult solutions.md until you have completed or paused every part.

Question 1 — total differentiation (11 marks; 24 minutes)

Let

$$c(p, w) = \frac{p^2}{w} + 2 \log w, \qquad p > 0, \quad w > 0.$$

W05-Q01-PA (M7; 5 marks; 10 minutes) Calculate the column gradient $\nabla c(p, w)$ and write the total differential. At $(p, w) = (2, 1)$, use it to approximate the change in c when $\mathrm{d}p = 0.02$ and $\mathrm{d}w = -0.01$.

W05-Q01-PB (M7; 6 marks; 14 minutes) Define

$$z(\theta) = x(\theta)^2 y(\theta) + \theta \log y(\theta), \qquad x(\theta) = 1 + \theta, \quad y(\theta) = 2 - \theta.$$

State the domain condition near $\theta = 0$, write the total-derivative formula including the direct effect through θ, and calculate $\mathrm{d}z/\mathrm{d}\theta$ at $\theta = 0$.

Question 2 — implicit differentiation (12 marks; 29 minutes)

W05-Q02-PA (M7; 5 marks; 12 minutes) The equation

$$F(x, \theta) = x^2 + \theta x - 6 = 0$$

holds at $(x, \theta) = (2, 1)$. State the condition that allows the equation to define $x(\theta)$ locally, verify it at the point, and find $\mathrm{d}x/\mathrm{d}\theta$ there.

W05-Q02-PB (M7; 7 marks; 17 minutes) Let $\mathbf{x}(\theta) = [x(\theta), y(\theta)]^{\mathsf{T}}$ be defined locally by

$$\begin{aligned} F_1(x, y, \theta) &= x + y + \theta - 3 = 0, \\ F_2(x, y, \theta) &= x^2 + y^2 - \theta - 5 = 0. \end{aligned}$$

At $(x, y, \theta) = (1, 2, 0)$, state the dimensions of the relevant Jacobian, verify its nonsingularity, and calculate $\mathrm{d}\mathbf{x}/\mathrm{d}\theta$.

Question 3 — homogeneity and Euler's theorem (10 marks; 24 minutes)

W05-Q03-PA (M7; 10 marks; 24 minutes) On $x > 0, y > 0$, let

$$f(x, y) = x^{1/3} y^{2/3}.$$

Identify its degree of homogeneity and verify Euler's theorem directly. A student then says that $r(x, y) = x + y^2$ is homogeneous because each term is a power. Show that the claim is false. Finally, show that

$$h(x,y)=\frac{x^2}{x+y}, \qquad x>0, \quad y>0,$$

is homogeneous, identify its degree, and verify Euler's identity at $(x,y)=(2,1)$.

Question 4 — quadratic forms and definiteness (12 marks; 28 minutes)

`W05-Q04-PA` **(M8; 5 marks; 12 minutes)** Let

$$\mathbf{A}=\begin{bmatrix}2 & 3\\ -1 & 2\end{bmatrix}, \qquad q(\mathbf{x})=\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}.$$

Find the symmetric matrix that represents the same quadratic form and classify q by definiteness. Justify the classification.

`W05-Q04-PB` **(M8; 7 marks; 16 minutes)** For $a\in\mathbb{R}$, classify the quadratic form associated with

$$\mathbf{B}(a)=\begin{bmatrix}2 & a\\ a & 2\end{bmatrix}$$

as positive definite, positive semidefinite or indefinite. Treat all values of a, including the boundary cases, and explain why no other definiteness class occurs.

Question 5 — convexity, concavity and quasiconcavity (12 marks; 28 minutes)

`W05-Q05-PA` **(M8; 6 marks; 13 minutes)** Determine whether each set is convex:

$$C=\{(x,y)\in\mathbb{R}^2 : y\geq x^2\}, \qquad D=\{(x,y)\in\mathbb{R}^2 : y\leq x^2\}.$$

Use the definition or a valid convexity result for C, and give an explicit counterexample for any set that is not convex.

`W05-Q05-PB` **(M8; 6 marks; 15 minutes)** On the convex domain $\mathbb{R}^2_{++}:=\{(x,y):x>0,y>0\}$, classify

$$g(x,y)=xy$$

as convex, concave and/or quasiconcave. Use the Hessian for convexity and concavity, and a separate upper-contour-set argument for quasiconcavity.

Question 6 — a discrete joint distribution (13 marks; 29 minutes)

The joint PMF of $X\in\{0,1\}$ and $Y\in\{0,1,2\}$ is

x	$p_{X,Y}(x,0)$	$p_{X,Y}(x,1)$	$p_{X,Y}(x,2)$
0	1/10	2/10	1/10
1	1/10	2/10	3/10

`W05-Q06-PA` **(S5; 6 marks; 13 minutes)** Derive both marginal PMFs, check that they are valid, and determine whether X and Y are independent.

W05-Q06-PB (S5; 7 marks; 16 minutes) Derive the conditional PMF of $X \mid Y = y$ for every $y \in \{0, 1, 2\}$. Hence find $\mathbb{E}[X \mid Y]$ and verify the law of iterated expectation.

Question 7 — a continuous joint distribution (9 marks; 27 minutes)

Suppose

$$f_{X,Y}(x, y) = c \quad \text{for } 0 < x < y < 1,$$

and $f_{X,Y}(x, y) = 0$ otherwise.

W05-Q07-PA (S5; 5 marks; 13 minutes) Find c, derive $f_X(x)$ and $f_Y(y)$ with their supports, and derive $f_{X|Y}(x \mid y)$ with all conditions stated.

W05-Q07-PB (S5; 4 marks; 14 minutes) Calculate $\mathbb{E}[X \mid Y = y]$. Use iterated expectation to find $\mathbb{E}[X]$, then verify the result directly from the joint density.

Question 8 — total variance (10 marks; 25 minutes)

W05-Q08-PA (S5; 10 marks; 25 minutes) Let $Y \sim \text{Bernoulli}(1/3)$. Conditional on $Y = 0$, $X \sim \text{Bernoulli}(1/4)$; conditional on $Y = 1$, $X \sim \text{Bernoulli}(3/4)$.

Use the law of total variance to calculate $\text{Var}(X)$, showing both components separately. Then derive the marginal distribution of X and use it to verify your answer.

Question 9 — retrieval: eigenstructure (4 marks; 10 minutes)

Complete this part without consulting earlier notes.

W05-Q09-PA (M5; 4 marks; 10 minutes) For

$$\mathbf{K} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

find the eigenvalues and a basis of eigenvectors for each eigenspace. State whether $\mathbf{K}$ is diagonalizable and give a valid diagonalisation.

Question 10 — retrieval: conditioning and moments (7 marks; 16 minutes)

Complete both parts without consulting earlier notes.

W05-Q10-PA (S2; 4 marks; 8 minutes) Let A be an event with $\mathbb{P}(A) = 1/4$. Suppose

$$\mathbb{P}(C \mid A) = \frac{3}{4}, \qquad \mathbb{P}(C \mid A^c) = \frac{1}{4}.$$

Calculate $\mathbb{P}(C)$ and $\mathbb{P}(A \mid C)$, stating the rule used.

W05-Q10-PB (S4; 3 marks; 8 minutes) Suppose $\mathbb{E}[U] = 2$ and $\text{Var}(U) = 3$. For $V = 4 - 2U$, calculate $\mathbb{E}[V]$ and $\text{Var}(V)$. Explain why the additive constant affects one quantity but not the other.

Self-test questions

Set aside 25 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed. State the condition, support or identity that validates each method. The self-test has 20 marks and does not contribute to a checkpoint or mock-examination score.

Questions

`W05-ST01-PA` — scalar implicit differentiation (`M7`; 5 marks; 6 minutes) The equation

$$F(x, \theta) = xe^x + \theta - 1 = 0$$

holds at $(x, \theta) = (0, 1)$. Verify the local implicit-function condition and calculate $\mathrm{d}x/\mathrm{d}\theta$ at the point.

`W05-ST02-PA` — definiteness and curvature (`M8`; 5 marks; 6 minutes) Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \qquad q(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}.$$

Classify q by definiteness. Then decide whether q, viewed as a function on $\mathbb{R}^2$, is convex or concave. Justify both conclusions.

`W05-ST03-PA` — conditional expectation (`S5`; 5 marks; 6 minutes) The joint PMF of $X, Y \in \{0, 1\}$ is

x	$p_{X,Y}(x, 0)$	$p_{X,Y}(x, 1)$
0	1/3	1/6
1	1/6	1/3

Find $\mathbb{E}[X \mid Y]$, then verify the law of iterated expectation.

`W05-ST04-PA` — total expectation and variance (`S5`; 5 marks; 7 minutes) Let $Y \sim \text{Bernoulli}(1/2)$. Suppose

$$\mathbb{E}[X \mid Y] = 1 + Y, \qquad \mathrm{Var}(X \mid Y) = 2.$$

Use iterated expectation and total variance to calculate $\mathbb{E}[X]$ and $\mathrm{Var}(X)$, showing the two variance components separately.